

(1.1)

$$\text{Dikaz } \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2).$$

Upravíme výrazy v pravce:

$$\begin{aligned} \|\vec{u} + \vec{v}\|^2 &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

$$\|\vec{u} - \vec{v}\|^2 = \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 - \langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

Dostáváme tedy:

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) - \|\vec{u}\|^2 - \|\vec{v}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

Takže platí:

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) \Rightarrow \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$\text{Dikaz } \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2).$$

Upravíme výrazy v pravce:

$$\begin{aligned} \|\vec{u} + i\vec{v}\|^2 &= \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 + i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + i(2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)) = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

$$\begin{aligned} \|\vec{u} - i\vec{v}\|^2 &= \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} \rangle + \langle -i\vec{v}, -i\vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 - i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 - i(2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)) = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

Dostáváme tedy:

$$\begin{aligned} \|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) - \|\vec{u}\|^2 - \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \\ &= 4\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

Takže platí:

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = 4\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \Rightarrow \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

Při úpravách jsme využili slovo-symetrii a lineární a úpravy s komplexním číslům.